

EAS 6134 Inverse Methods in EAS
Homework 6: A nonlinear parameter estimation problem

1. The Arrhenius relationship for thermally activated semi-conduction in minerals is $\sigma(t) = \sigma_0 e^{-A/kt}$ where $\sigma(t)$ is the electrical conductivity at temperature t in degrees Kelvin, k is Boltzmann's constant and A is the activation energy. This has been used to model the electrical conductivity data for the mineral olivine as a function of temperature. Olivine is a major constituent of Earth's mantle. Dunite is an ultramafic plutonic rock that is composed almost exclusively of olivine.

- Download and plot the data for the Jackson County dunite in `jcd.dat`. The file contains a header line followed by 68 data in 2 columns, temperature ($^{\circ}C$) and conductivity (S/m).

- Explain how to perform an appropriately parametrized linear least squares fit to find a single activation energy A and the constant σ_0 .

- Using Matlab, Python, (or otherwise if you prefer), and omitting the first 14 data in the file, find estimates for σ_0 and A .

- If the errors in the conductivity measurements are 5%, what are the uncertainties in your derived parameters? Use an appropriate analysis of residuals to make a suitable argument as to whether 5% seems like a reasonable estimate of the data uncertainty.

- Redo the analysis with the entire data set and additional parameters that might allow a better fit. For example, you could add another exponential term $\sigma_1 e^{-A_1/kt}$ or if a different conduction mechanism (doping) is also involved perhaps add a constant to the first exponent, or to both. Comment on the differences. Modify your modelling approach to accommodate the additional data and solve via a nonlinear least squares method of your choice (e.g., Levenberg-Marquardt, or MC method.)

- Outline a strategy for Bayesian/trans-dimensional inversion that would allow for uncertainty quantification in your parameter estimates.